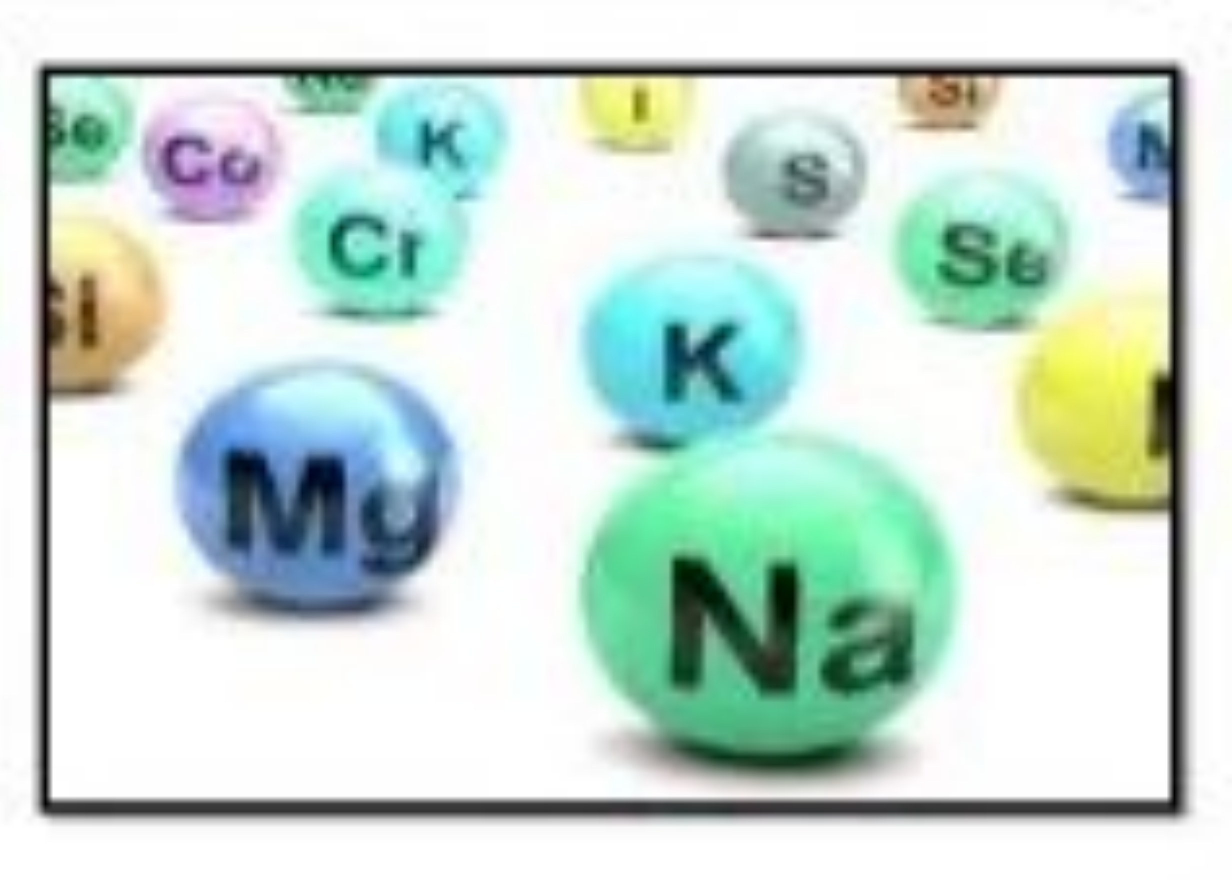
SODIUM, POTASSIUM & CHLORIDE

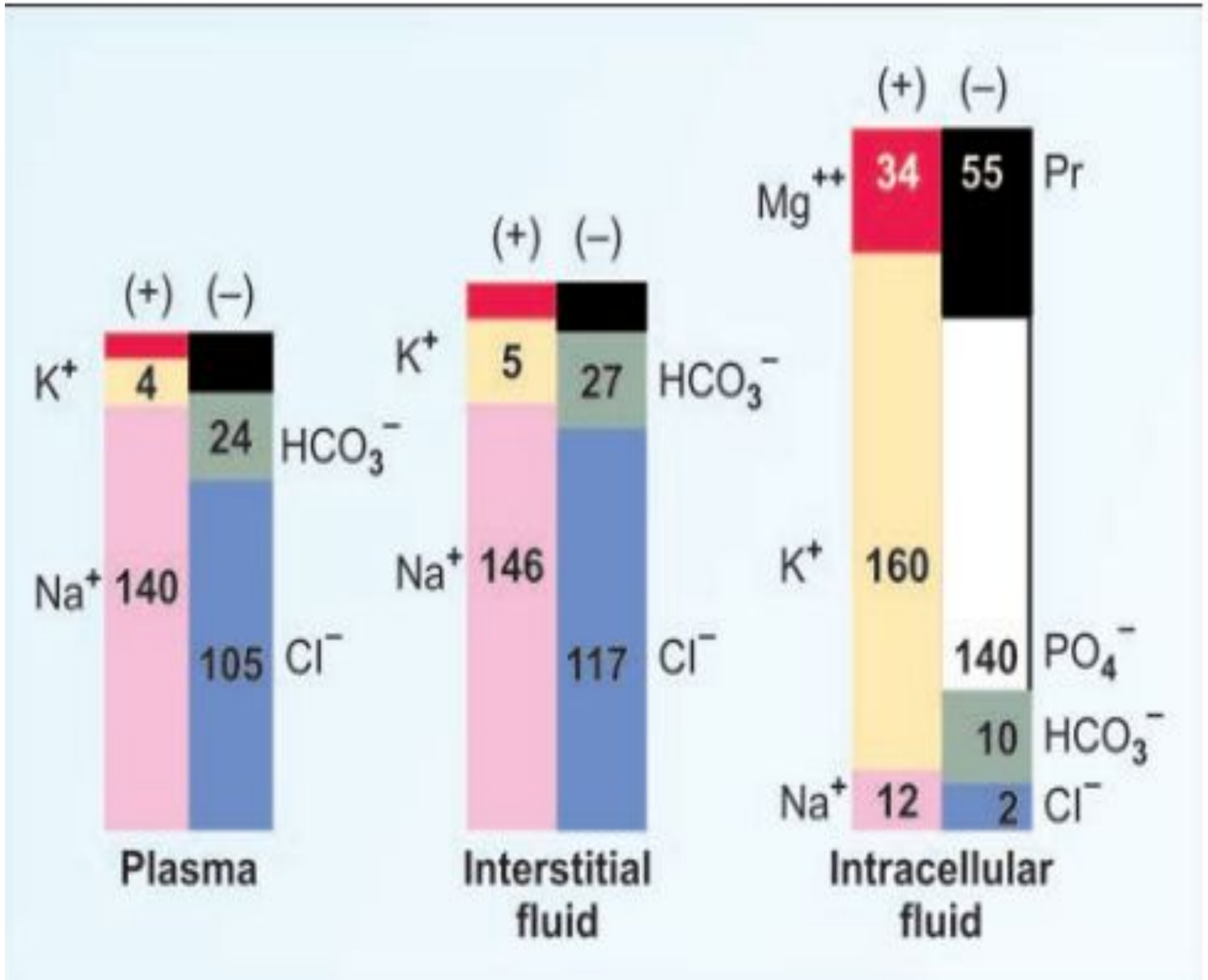


Dr. Humaira Aman Ali

Electrolyte concentration of body fluid compartments

Solutes	Plasma mEq/L	Interstitial fluid (mEq/L)	Intracellular fluid (mEq/L)
Cations:			
Sodium	140	146	12
Potassium	4	5	160
Calcium	5	3	–
Magnesium	1.5	1	34
Anions:			
Chloride	105	117	2
Bicarbonate	24	27	10
Sulphate	1	1	–
Phosphate	2	2	140
Protein	15	7	54
Other anions	13	1	–

Gamblegrams showing composition of fluid compartments



SODIUM (Na+)

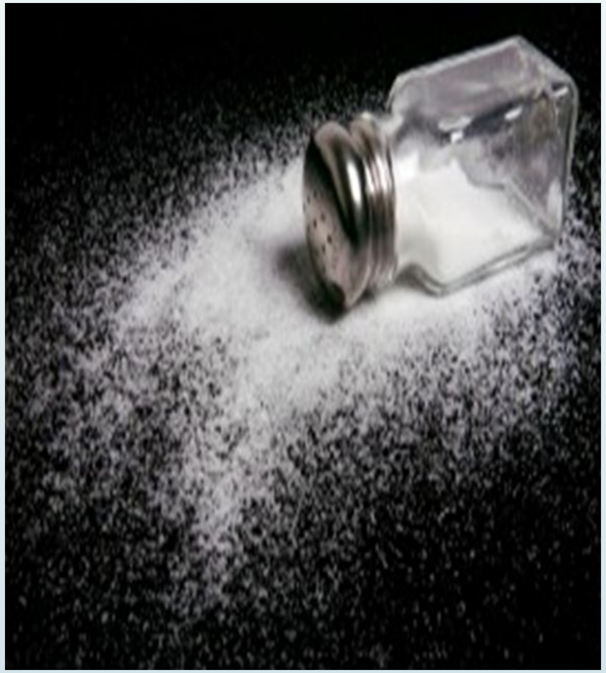
- Sodium is the chief electrolyte which is found in large conc. in extracellular fluid compartment and is one of the most abundant element on Earth.

- The sodium is found in the body mainly associated with chloride as NaCl and $NaHCO_3$.

- ***Sources:*** Sodium is widely distributed in food material. However, major source is **table-salt** used in cooking or seasoning. It is also found in **cheese**, **butter, processed food, salted nuts, sauces and olives.**

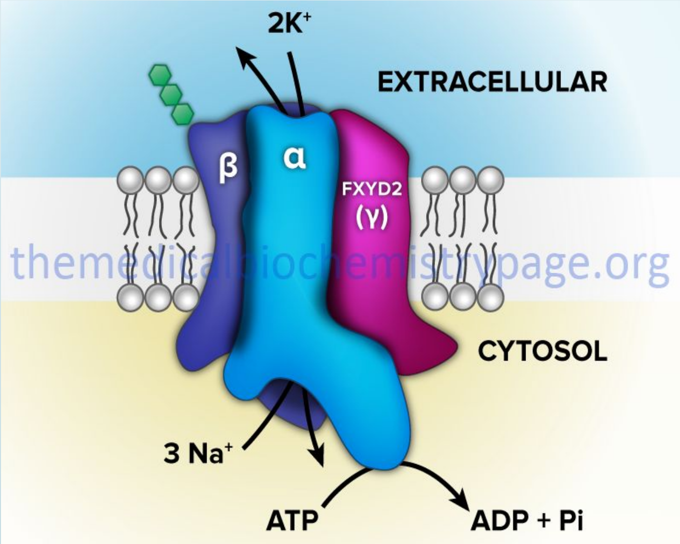
Daily Requirement:

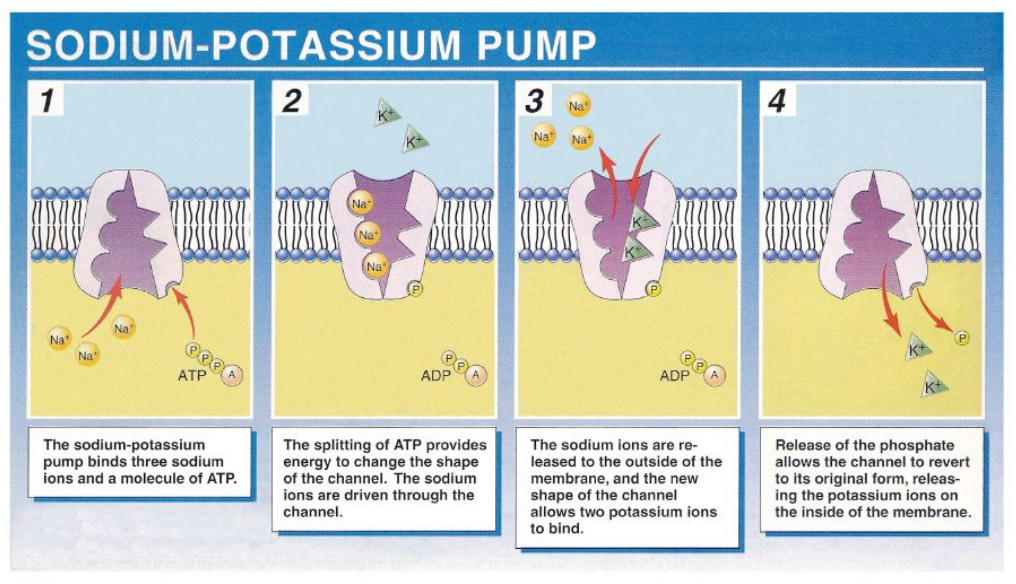
- According to U.S. Food and Drug Administration (FDA), the daily value for sodium is less than **2,300 milligrams (mg)** per day for a healthy adult.
- The American Heart Association (AHA) recommends no more than 2,300 milligrams (mg) a day and moving toward **an ideal limit of no more than 1,500 mg per day for most adults**.
- Reducing to 1,000 milligrams a day can significantly improve blood pressure and heart health.



SODIUM PUMP

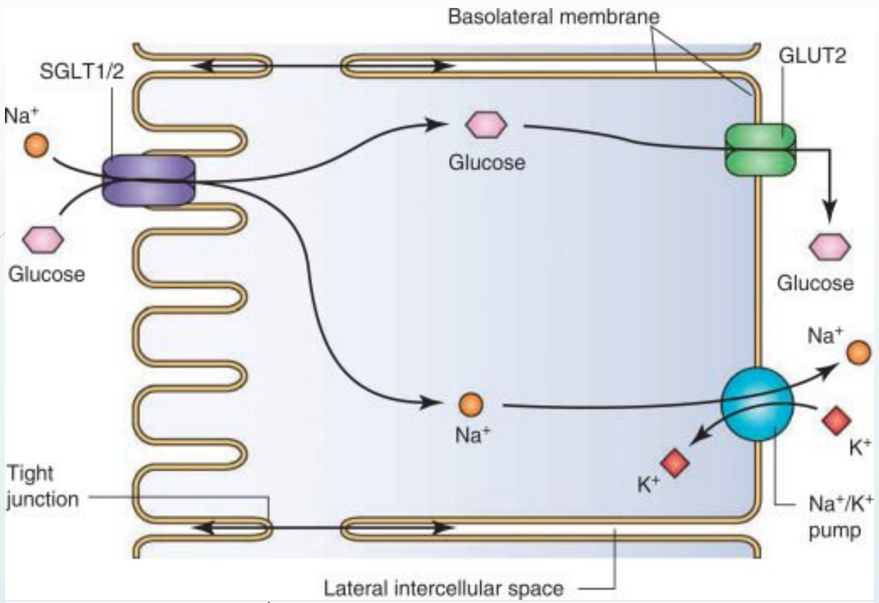
- This is also called as Na+-K+ ATPase. Na-pump is an enzyme, Na+-K+-ATPase.
- The enzyme hydrolyses a high energy phosphate bond of ATP and uses the energy thus released to transport three Na+ ions outside and simultaneously two K+ ions inside across the cell membrane.
- It is a glycoprotein composed of α and β chains. Its activity depends on presence of Na+ and K+ and requires ATP and Mg++ ions as cofactor.
- The Na-pump is very active in those cells where activities depend largely on transmembrane Na+ fluxes, e.g. nervous, muscle fibers, renal tubules cells, intestinal mucosal cells.

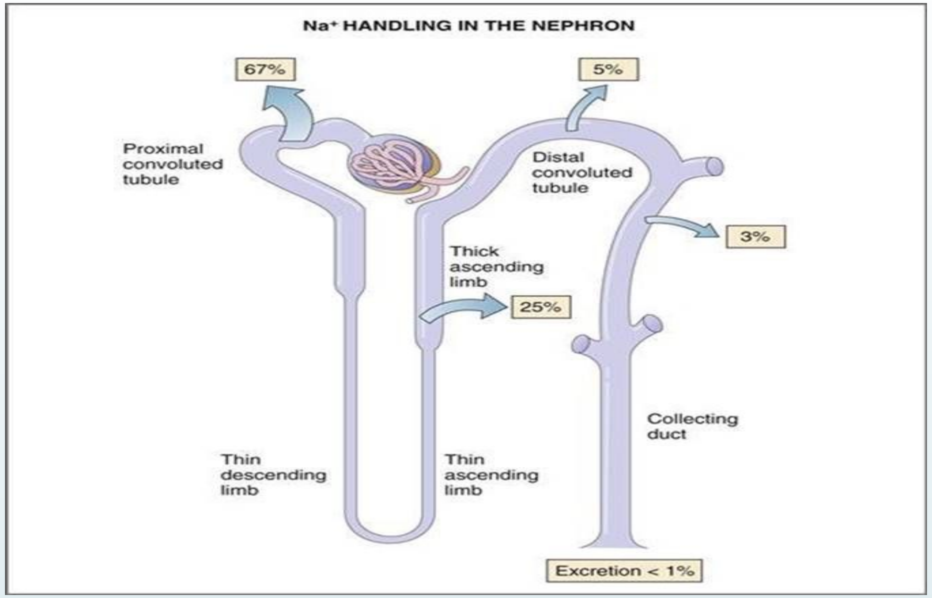


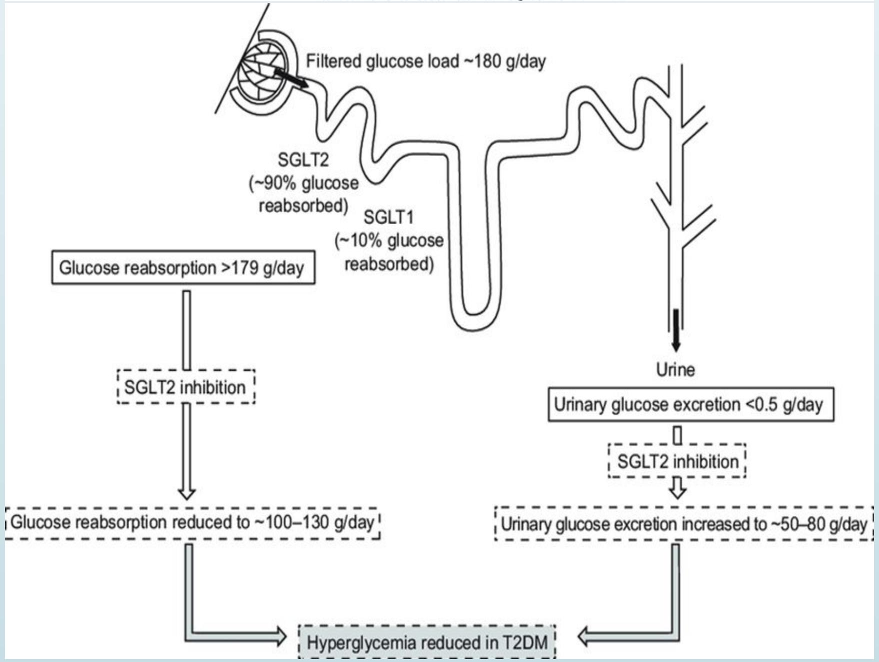


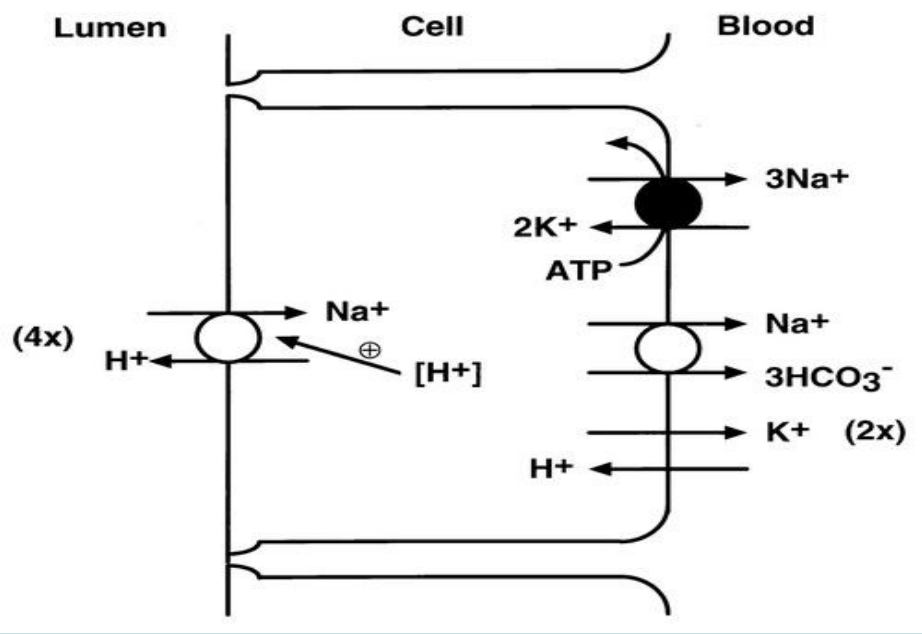
Renal sodium reabsorption and Excretion of Na:

- Every 24 hours approximately, 25000 mmol of sodium are filtered by the kidneys.
- However, due to tubular reabsorption less than 1 per cent of this sodium appears in the urine (100-200 mM/day).
- Approximately 70 per cent of the filtered sodium is reabsorbed in proximal tubule. Further 20–30 per cent of filtered Na^+ is reabsorbed by ascending loop of Henle.
- Renal sodium reabsorption uses Na-H antiport, Na-glucose symport, sodium ion channels (minor). It is stimulated by angiotensin II and aldosterone.


Basolateral membrane
SGLT1/2
GLUT2
Na+
Glucose
Glucose
Glucose
Na+
Na+
K+
K+
Tight junction
Na+/K+ pump
Lateral intercellular space

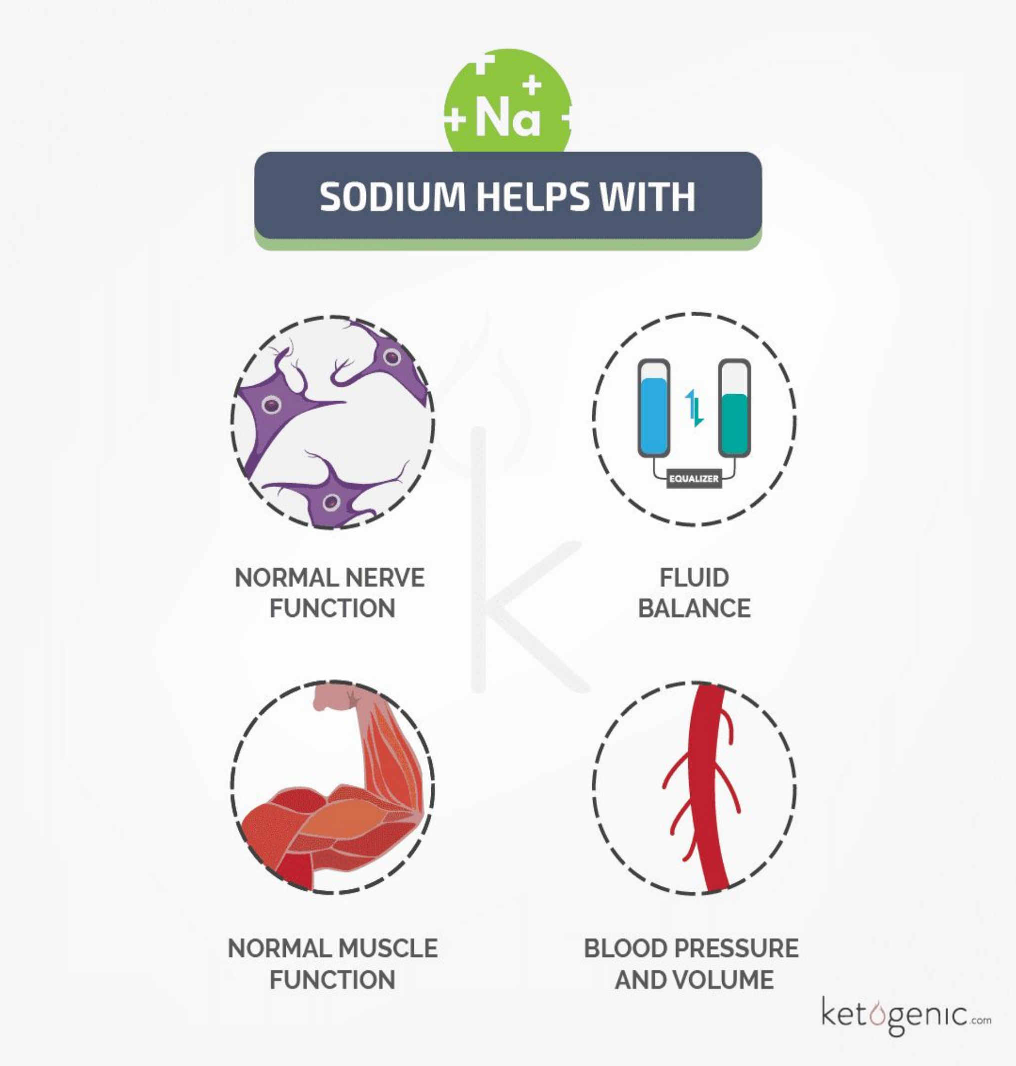

Na+ HANDLING IN THE NEPHRON
67%
5%
Proximal convoluted tubule
Distal convoluted tubule
3%
Thick ascending limb
25%
Collecting duct
Thin descending limb
Thin ascending limb
Excretion < 1%


Filtered glucose load ~180 g/day
SGLT2 (~90% glucose reabsorbed)
SGLT1 (~10% glucose reabsorbed)
Glucose reabsorption >179 g/day
SGLT2 inhibition
Urine
Urinary glucose excretion <0.5 g/day
SGLT2 inhibition
Glucose reabsorption reduced to ~100–130 g/day
Urinary glucose excretion increased to ~50–80 g/day
Hyperglycemia reduced in T2DM


Lumen
Cell
Blood
3Na+
2K+
ATP
(4x)
Na+
H+
⊕
[H+]
Na+
3HCO3-
K+ (2x)
H+

FUNCTIONS

- Maintenance of osmotic pressure and fluid balance: Sodium maintains osmotic pressure of extracellular fluids and helps in retaining water in ECF.
- Neuromuscular excitability.
- Regulates acid-base balance in association with chloride and bicarbonates.
- Role in Action Potential: A local depolarization of nerve or muscle fiber is observed in stimulation. This rapidly increases its permeability to Na+ causing considerable transmembrane influx of Na+.



Hypernatraemia

- Hypernatremia is defined as a rise in serum sodium concentration to a value exceeding 145 mmol/L.
- Causes:
  - Primary hyperaldosteronism (Conn’s syndrome): e:g aldosterone producing adenoma
  - Cushing's disease
  - Prolonged steroid therapy
  - Excessive intake of salt
  - In dehydration, when water is predominantly lost, blood volume is decreased with apparent increased concentration of sodium

cont..

- Elderly patients with poor water intake, and inability to express thirst
- Drugs: Anabolic steroids Oral contraceptives, Loop diuretics and Osmotic diuretics
- In pregnancy, steroid hormones cause sodium retention in the body
- Diabetes insipidus: These diseases are characterized by lack of antidiuretic hormone (ADH) or failure of the hormone to act on its target cells.

Hyponatraemia

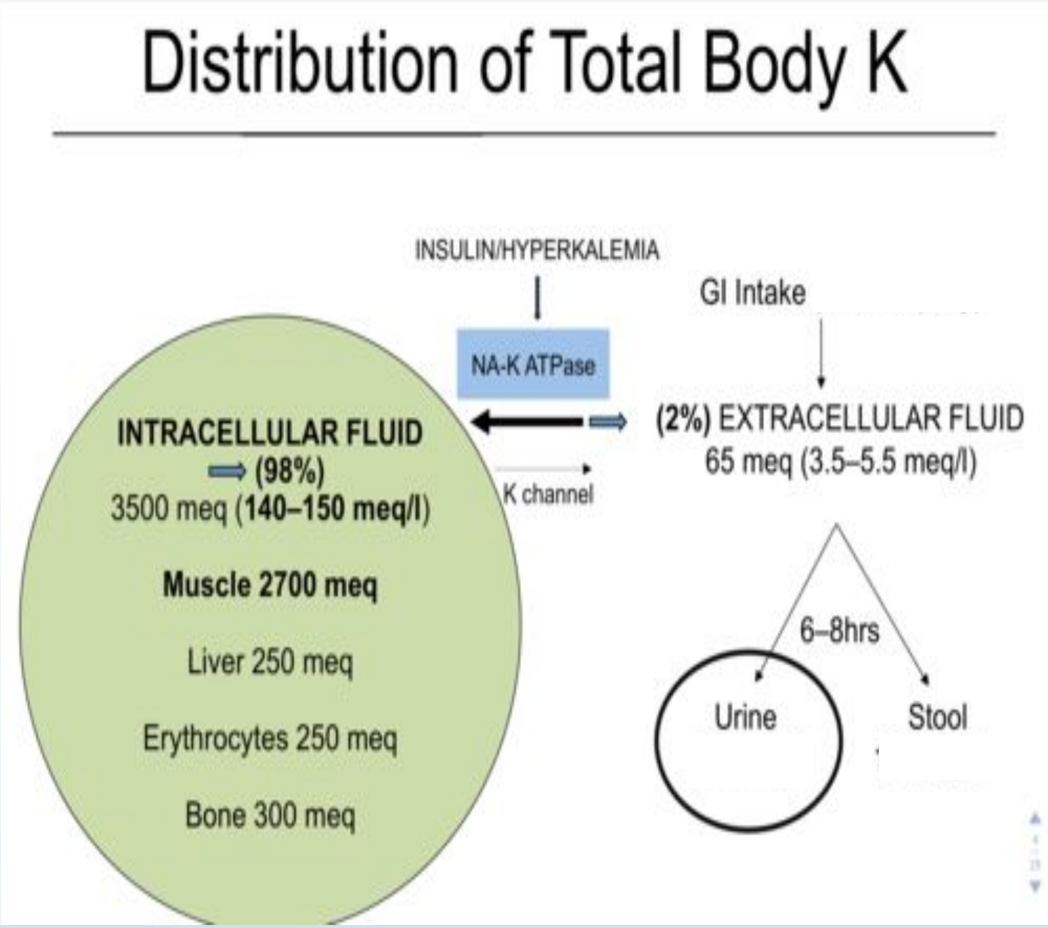
- Decreased sodium level in blood is called hyponatremia.
- Causes:
  - Vomiting, diarrhea and excessive sweating
  - Burns result in the **loss of both water and sodium**
  - Addison's disease (adrenal insufficiency)
  - Renal tubular acidosis (tubular reabsorption of sodium is defective)
  - Chronic renal failure: due to kidney dysfunction, Na+ is not reabsorbed

cont..

- Congestive cardiac failure b/c of Diuretics administration and in advance stages.
- Hyperglycemia and ketoacidosis: Hyperglycemia causes hyperosmolality, and the water moves from intracellular space to extracellular space, which in turn produces a dilutional decrease in serum sodium level
- SIADH: Syndrome of inappropriate secretion of anti-diuretic hormone
- Drugs: ACE inhibitors, Vasopressin, use of diuretics.

POTASSIUM (K+)

- Potassium is the major intracellular cation.
- Over 98% is in the intracellular compartment (primarily in the muscle, subcutaneous tissue, and red blood cells) and 2% is in the extracellular compartment.
- The intracellular concentration gradient is maintained by the Na+-K+ ATPase pump.



REQUIREMENT, SOURCES, NORMAL PLASMA LEVEL, EXCRETION and REGULATION

- **Requirement:**
  - Potassium requirement is 3-4 g per day.
- **Sources:**
  - Sources rich in potassium, but low in sodium are **banana, orange, apple, pineapple, almond, dates, beans, yam and potato. Coconut water is a very good source of potassium.**
- **Normal plasma level:**
  - The normal conc. of plasma potassium is 3.5-5 mEq/L.
- **Excretion:**
  - 90% of excess potassium is excreted through kidneys in the urine.
  - The rest 10% potassium is also excreted in gastrointestinal tract, saliva, gastric juice, bile, pancreatic and intestinal juices.

Regulation

- The majority of the filtered K+ is reabsorbed in the proximal tubule.
- The control of secretion occurs in the cortical collecting duct. The exchange of potassium for sodium at the renal tubules is a mechanism to conserve sodium and excrete potassium.
- This is controlled by aldosterone. Aldosterone and corticosteroids increase the excretion of K+.
- On the other hand, K+ depletion will inhibit aldosterone secretion.

Note:

- **Insulin promotes the entry of K^+ into skeletal muscles and hepatic cells.**

FUNCTIONS

- Many functions of potassium and sodium are carried out in coordination with each other.
  - Nerve conduction
  - Muscular function
  - Osmotic pressure: regulates intracellular osmolality
  - Protein synthesis: K^+ and Mg^{2+} both ions contribute to the stability of various RNA structures
  - Acid-base balance
  - It has an important role in cardiac function

Hypokalemia

- Hypokalemia is generally defined as a serum potassium level of less than 3.5 mEq/L (3.5 mmol/L).
- Causes:
  - Cushing's syndrome
  - Hyperaldosteronism
  - Hyper reninism, renal artery stenosis
  - 17 alpha hydroxylase deficiency and 11 beta hydroxylase deficiency (genetic disorder of steroid biosynthesis that causes decreased production of glucocorticoids and increased synthesis of mineralocorticoid precursors)

cont..

- Alkalosis
- Insulin therapy
- Gastrointestinal loss: Diarrhea, vomiting
- Deficient intake or low potassium diet
- Malabsorption
- Intravenous saline infusion in excess
- Drugs Insulin Salbutamide Osmotic diuretics Thiazides, acetazolamide Corticosteroids

Hyperkalemia

- Plasma potassium level above 5.5 mmol/L is known as hyperkalemia. Since the normal level of K+ is kept at a very narrow margin, even minor increase is life-threatening.
- There is increased membrane excitability, leads to ventricular arrhythmia and ventricular fibrillation.
- Lowering cell-resting action potential and prevention of repolarization, leading to flaccid muscle paralysis, bradycardia and cardiac arrest.
- ECG shows elevated T wave, widening of QRS complex and lengthening of PR interval.

Causes

- Decreased renal excretion of potassium: Obstruction of urinary tract, renal failure, severe volume depletion (heart failure)

- Entry of potassium to extracellular space: Increased hemolysis, tissue necrosis, burns tumor lysis after chemotherapy, rhabdomyolysis, crush injury, excess potassium supplementation

- Metabolic acidosis

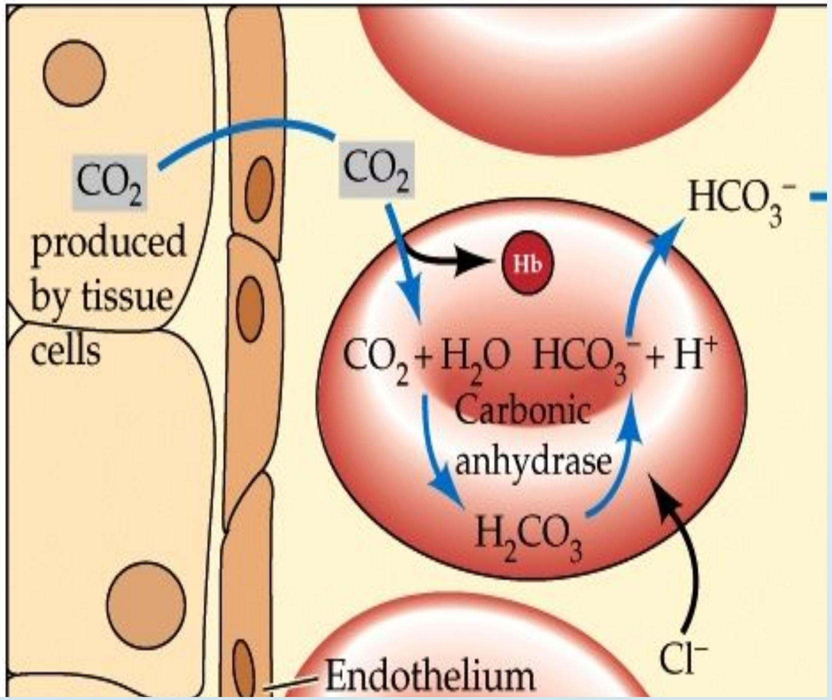
- Insulin deficiency (diabetes mellitus)

cont..

- Tissue hypoxia
- Improper blood collection (hemolysis)
- Thrombocytosis
- Leukocytosis
- Hyperkalemic periodic paralysis
- Drugs Spiranolactone ACE inhibitors Beta blockers

CHLORIDE (Cl–)

- **Functions:**
  - Chloride is important in the formation of hydrochloric acid in gastric juice.
  - Chloride ions are also involved in chloride shift.
- **Sources:**
  - Table salt
  - Vegetables and meats have small proportions of chloride.
  - It is also available in the ‘chlorinated water’ which is normally supplied as a process of purification of water for drinking purpose.
- **Daily Requirement:** About 100-200 mmol is taken in diet as sodium chloride (table salt)



CHLORIDE SHIFT

- **Chloride concentration in plasma** is 96-106 mEq/L and in CSF, it is about 125 mEq/L.
- Chloride concentration in CSF is higher than any other body fluids.
- **Distribution:**
  - Plasma
  - CSF
  - Cells
  - Muscles

- **Excretion**
  - Sweat
  - Faeces
  - Renal

- 99 per cent of the Cl– in the glomerular filtrate is reabsorbed by renal tubules mainly in proximal tubule (60-70%) and then in ascending loop of Henle (20-25%) followed by distal tubule, collecting duct (10-15%).

- **Regulations:**
  - Similar to that of sodium.
  - Increase in blood volume decreases reabsorption of chloride and vice-versa.
  - Plasma levels of chloride vary with and to a great extent depend on the plasma conc. of Na and HCO_3^-.

↓ Na associated with Cl^- ↓
↑ Na usually associated with Cl^- ↑
↑ HCO_3^- associated with Cl^- ↓
↓ HCO_3^- associated with Cl^- ↑

THANK YOU